

An Investigation into Music Habits at Doral Academy



AP Statistics ASA Project 2025-2026

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Introduction & Research Question

The Problem:

In high-pressure academic environments, students often look for ways to optimize their study time. One common strategy is listening to music, but its effect on deep focus remains unclear.

Primary Research Question:

Does listening to music while studying affect students' perceived concentration or productivity?

Secondary Variables Examined:

- **Music type:** Instrumental vs. lyrical
- **Listening method:** Headphones vs. speakers

Hypotheses & Variables

Null Hypothesis (H0): The mean perceived concentration and productivity scores are the same for students who study with music and students who study in silence

$(\mu_{\text{music}} = \mu_{\text{silence}})$.

Alternative Hypothesis (Ha): The mean perceived concentration and/or productivity scores differ between students who study with music and students who study in silence

$(\mu_{\text{music}} \neq \mu_{\text{silence}})$.

Operational Definitions:

- **Independent Variable:** Study environment (Silence/Instrumental/Lyrical).
- **Dependent Variable:** 1–5 Likert scale for Concentration and Productivity.

Methodology

Population:

Students at Doral Academy

Sampling Method:

Stratified random sampling by grade level. Two classes from 1 teacher that included only juniors and seniors were **randomly selected**, ensuring representation from both grade levels.

Sample Size:

$n = 57$ students (24 juniors, 33 seniors).

Data Collection:

An anonymous Google Form survey was used to reduce social desirability bias and encourage honest responses.

Independence:

The sample represented less than 10% of the total upperclassman population, satisfying the independence condition.

How do you usually listen to your music? *

- ☐ Headphones (Noise-canceling)
- ☐ Headphones (Standard)
- ☐ Open speakers

On average, how many hours of homework/studying do you do per night? *

- ☐ 0-1 hours
- ☐ 1-2 hours
- ☐ 2-3 hours
- ☐ 3+ hours

What type of music do you listen to most often while working? *

- ☐ None (I prefer silence)
- ☐ Instrumental (Lo-fi, Classical, Soundtracks)
- ☐ Pop / Rap (Music with lyrics)
- ☐ Rock / Metal
- ☐ Other: _____

Do you find yourself singing along or paying attention to lyrics rather than your schoolwork? *

- ☐ Always
- ☐ Frequently
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

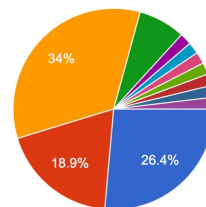
On a scale of 1–5, how would you rate your concentration while studying with your preferred background noise? *

Very Distracted 1 2 3 4 5 Deeply Focused

On a scale of 1–5, how productive do you feel when you have music playing? *

Not productive at all 1 2 3 4 5 Extremely Productive

What type of music do you listen to most often while working?



- None (I prefer silence)
- Instrumental (Lo-fi, Classical, Soundtracks)
- Pop / Rap (Music with lyrics)
- Rock / Metal
- Rock, Hip Hop, and Rap
- Latin
- Mix of all
- All (except country)

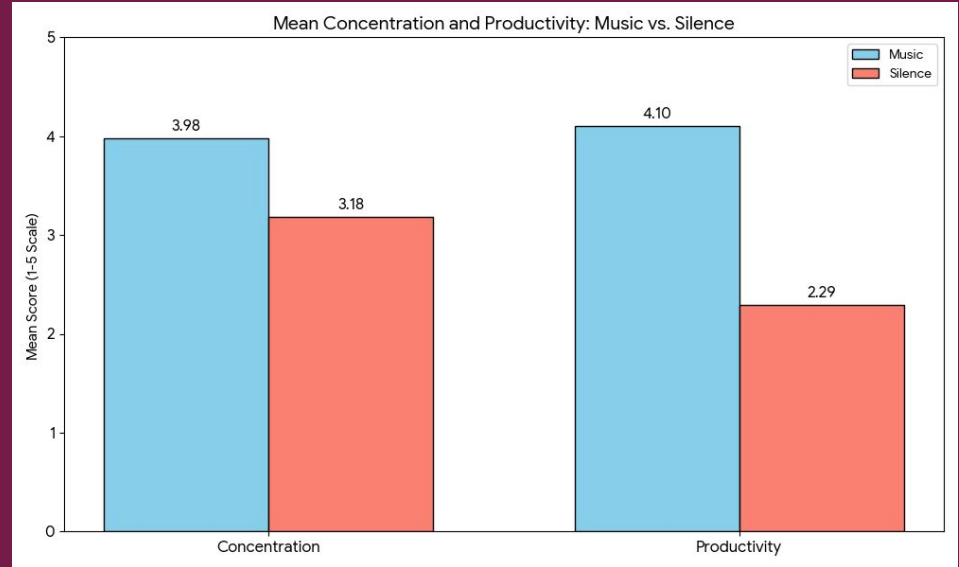
Data Visualized

Concentration Means: Music (3.98) vs. Silence (3.18).

Productivity Means: Music (4.10) vs. Silence (2.29).

Students who studied with music reported a higher average **concentration score (3.98)**, compared to students who studied in silence (**3.18**). This means students using music felt more focused overall.

For productivity, students studying with music reported a higher average score (**4.10**), while students studying in silence reported **2.29**. This shows a much larger difference, suggesting that students felt significantly more productive when music was part of their study environment.



Inference: Concentration (t-test)

Test:

Two-sample t-test for a difference in mean concentration between students who study with music and those who study in silence. A two-sample t-test was used because we were comparing the mean concentration of two independent groups: students who studied with music and students who studied in silence.

Conditions:

Randomization through stratified sampling, the 10% condition- 56 < 0,1(all Juniors and Seniors in Doral Academy High School), and approximate normality were met

Results:

$t \approx 2.96$, $df = 56$

$p\text{-value} \approx 0.005$

Decision & Conclusion:

Since the p-value is less than 0.05, we reject the null hypothesis. There is **statistical evidence of a difference in mean perceived concentration** between students who study with music and those who study in silence, with higher average concentration reported by music listeners.

Inference: Productivity (Confidence Interval)

95% Confidence Interval:

Difference in means $\pm t^*SE \rightarrow (3.98 - 3.18) \pm 2.003*0.41$

[1.00, 2.64] points

Interpretation:

We are 95% confident that the true difference in mean productivity (music – silence) for juniors and seniors at Doral Academy is between **1.00 and 2.64 points higher** for students who study with music.

Because the entire interval is above 0, this provides statistical evidence that music is associated with higher perceived productivity.

Practical Significance:

An increase of nearly 2 points on a 5-point Likert scale represents a substantial shift in perceived study efficiency, suggesting the effect is not only statistically significant but also practically meaningful.

ANOVA: Instrumental vs. Lyrical

Analysis of Variance (ANOVA) was used to examine whether mean productivity differs by music type. This test is appropriate because productivity was analyzed across multiple music categories rather than a single two-group comparison.

Instrumental music reported a higher mean productivity score ($\bar{x} \approx 4.30$) than lyrical music ($\bar{x} \approx 3.82$).

The results produced an F-statistic (ratio comparing the variance between group means to the variance within the groups) of 4.2 with a p-value of 0.045. Since the p-value < 0.05 , we reject the null hypothesis and conclude that **productivity differs based on music type**. Instrumental music was associated with the highest reported productivity, likely due to reduced verbal interference.

Limitations & Discussion

Self-Reporting:

All the data was self-reported using a survey, meaning the results reflect **students' perceptions of concentration and productivity**, not objective measures such as test scores or GPA.

Observational Design:

Because this study was observational, **cause-and-effect conclusions cannot be made**. While music is associated with higher reported productivity, the results do not prove that listening to music directly causes improved focus.

Generalizability:

The sample included only **juniors and seniors at Doral Academy**, which limits how well the findings can be generalized to other schools, grade levels, or student populations.

Conclusion & Future Research

Key Takeaway:

For students at Doral Academy, the **study environment is significantly associated with perceived concentration and productivity**.

Recommendation:

Based on the results, students may benefit from **experimenting with instrumental music** while studying, as it was associated with the highest reported productivity.

Future Research:

Future studies could strengthen these findings by:

- Using an **experimental design** that assigns students to specific study environments
- Tracking **objective academic outcomes** such as GPA or test scores over time
- Expanding the sample to include **freshmen and sophomores** or students from other schools